

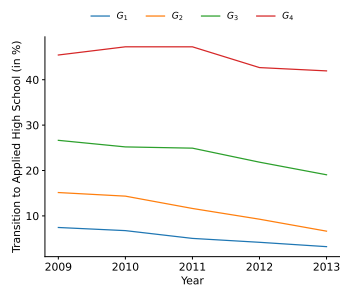
Nonstandard Educational Careers and the Returns to Vocational Education - Supplemental Appendix

July 13, 2026

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S1 Additional figures



(a) Applied High School Enrollment

Figure S1: Enrollment Rates Between 2009 and 2013

Note: This figure shows enrollment at applied high school for cohorts graduating from vocational school between 2009 and 2013 for each quartile of grades at the end of vocational school.

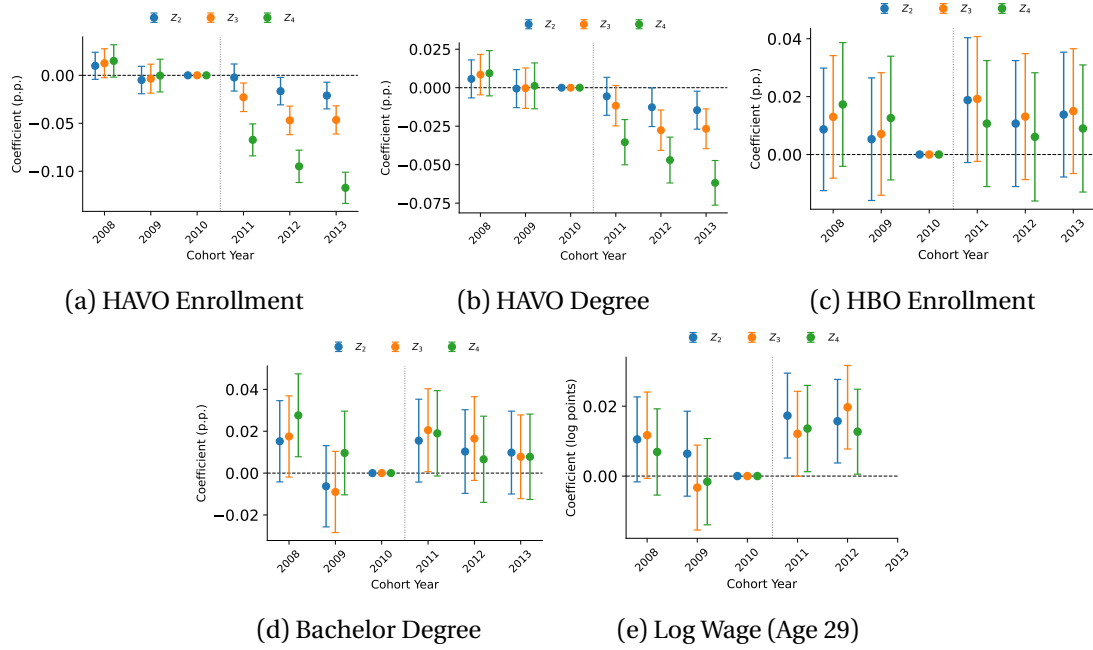


Figure S2: Event-study estimates using Z_1 alone as the control group

Note: Same event-study specifications as the main paper's Figure 3 (enrollment and graduation outcomes) and Appendix Figure A.1 (log wages) - i.e., the difference-in-differences specification in Equation 12 of the main paper - but Z_2 is no longer pooled with Z_1 into a combined control group; each panel shows Z_2 , Z_3 and Z_4 estimated separately against Z_1 (the lowest school type) alone as the sole control. Coefficients are relative to the 2010 reference cohort. Bars show 95% confidence intervals. The vertical dotted line marks the reform cutoff (2010/2011).

S2 Reduced-form regression estimates

Table S1 reports the full regression underlying the event-study figures in Figure S2 above, which uses Z_1 alone as the control group. Table S3 reports the full regression underlying the in-paper event-study figures, which pool Z_1 and Z_2 as the control group. Standard errors are in parentheses; */**/** denote significance at the 10%/5%/1% level.

Table S1: Full regression estimates: Z_1 alone as control

Variable	HAVO enrollment	HAVO graduation	Applied university enrollment	Bachelor degree
Intercept	0.1347*** (0.0038)	0.0750*** (0.0029)	0.5059*** (0.0056)	0.3086*** (0.0051)
$\mathbb{1}\{t = 2008\}$	0.0017 (0.0054)	0.0004 (0.0041)	-0.0078 (0.0078)	-0.0251*** (0.0072)
$\mathbb{1}\{t = 2009\}$	0.0088 (0.0054)	0.0077* (0.0042)	-0.0015 (0.0078)	-0.0042 (0.0072)
$\mathbb{1}\{t = 2011\}$	-0.0048 (0.0054)	0.0014 (0.0042)	-0.0145* (0.0079)	-0.0097 (0.0073)
$\mathbb{1}\{t = 2012\}$	-0.0050 (0.0054)	0.0069 (0.0043)	-0.0194** (0.0080)	-0.0079 (0.0074)
$\mathbb{1}\{t = 2013\}$	-0.0194*** (0.0052)	0.0033 (0.0042)	-0.0270*** (0.0079)	-0.0084 (0.0073)
Z_1	0.0525*** (0.0056)	0.0354*** (0.0044)	0.0140* (0.0076)	-0.0057 (0.0071)
Z_2	0.0991*** (0.0059)	0.0638*** (0.0047)	0.0245*** (0.0077)	-0.0059 (0.0071)

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Variable	HAVO enrollment	HAVO graduation	Applied university enrollment	Bachelor degree
Z_3	0.2176*** (0.0065)	0.1348*** (0.0054)	0.0596*** (0.0078)	0.0159** (0.0073)
$\mathbb{1}\{t = 2008\}:Z_1$	0.0094 (0.0080)	0.0057 (0.0063)	0.0087 (0.0108)	0.0152 (0.0099)
$\mathbb{1}\{t = 2009\}:Z_1$	-0.0046 (0.0080)	-0.0006 (0.0063)	0.0053 (0.0108)	-0.0063 (0.0099)
$\mathbb{1}\{t = 2011\}:Z_1$	-0.0059 (0.0079)	-0.0056 (0.0063)	0.0188* (0.0110)	0.0155 (0.0101)
$\mathbb{1}\{t = 2012\}:Z_1$	-0.0179** (0.0079)	-0.0127** (0.0064)	0.0107 (0.0111)	0.0103 (0.0102)
$\mathbb{1}\{t = 2013\}:Z_1$	-0.0189** (0.0077)	-0.0146** (0.0063)	0.0138 (0.0110)	0.0098 (0.0101)
$\mathbb{1}\{t = 2008\}:Z_2$	0.0153* (0.0084)	0.0085 (0.0067)	0.0130 (0.0108)	0.0175* (0.0099)
$\mathbb{1}\{t = 2009\}:Z_2$	-0.0027 (0.0084)	-0.0003 (0.0067)	0.0071 (0.0108)	-0.0090 (0.0099)
$\mathbb{1}\{t = 2011\}:Z_2$	-0.0238*** (0.0083)	-0.0117* (0.0067)	0.0192* (0.0110)	0.0205** (0.0101)
$\mathbb{1}\{t = 2012\}:Z_2$	-0.0423*** (0.0082)	-0.0276*** (0.0067)	0.0131 (0.0111)	0.0165 (0.0102)
$\mathbb{1}\{t = 2013\}:Z_2$	-0.0477*** (0.0080)	-0.0267*** (0.0066)	0.0150 (0.0110)	0.0078 (0.0102)
$\mathbb{1}\{t = 2008\}:Z_3$	0.0212** (0.0091)	0.0094 (0.0075)	0.0173 (0.0109)	0.0276*** (0.0101)
$\mathbb{1}\{t = 2009\}:Z_3$	0.0105 (0.0092)	0.0012 (0.0076)	0.0126 (0.0109)	0.0096 (0.0102)
$\mathbb{1}\{t = 2011\}:Z_3$	-0.0658*** (0.0091)	-0.0354*** (0.0091)	0.0107 (0.0111)	0.0190* (0.0104)
$\mathbb{1}\{t = 2012\}:Z_3$	-0.0876*** (0.0091)	-0.0470*** (0.0076)	0.0061 (0.0113)	0.0066 (0.0105)
$\mathbb{1}\{t = 2013\}:Z_3$	-0.1113*** (0.0088)	-0.0618*** (0.0074)	0.0090 (0.0112)	0.0078 (0.0104)
R-squared	0.033	0.017	0.003	0.001
N	199488	199488	199488	199488

Table S3: Full regression estimates: Z_1 and Z_2 pooled as control

Variable	HAVO enrollment	HAVO graduation	Applied university enrollment	Bachelor degree
Intercept	0.1288*** (0.0026)	0.0936*** (0.0022)	0.5132*** (0.0038)	0.3056*** (0.0035)
$\mathbb{1}\{t = 2008\}$	0.0060 (0.0037)	0.0031 (0.0032)	-0.0034 (0.0054)	-0.0172*** (0.0049)
$\mathbb{1}\{t = 2009\}$	0.0082** (0.0037)	0.0069** (0.0032)	0.0011 (0.0054)	-0.0073 (0.0049)
$\mathbb{1}\{t = 2011\}$	-0.0030 (0.0037)	-0.0017 (0.0032)	-0.0048 (0.0055)	-0.0016 (0.0051)
$\mathbb{1}\{t = 2012\}$	-0.0050 (0.0037)	-0.0001 (0.0032)	-0.0141** (0.0055)	-0.0026 (0.0051)
$\mathbb{1}\{t = 2013\}$	-0.0166*** (0.0036)	-0.0046 (0.0032)	-0.0200*** (0.0055)	-0.0033 (0.0051)
Z_2	0.0704*** (0.0050)	0.0452*** (0.0043)	0.0171*** (0.0065)	-0.0029 (0.0060)

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Variable	HAVO enrollment	HAVO graduation	Applied university enrollment	Bachelor degree
Z_3	0.1915*** (0.0057)	0.1162*** (0.0050)	0.0523*** (0.0067)	0.0189*** (0.0062)
$\mathbb{1}\{t = 2008\}:Z_2$	0.0079 (0.0071)	0.0058 (0.0061)	0.0087 (0.0092)	0.0096 (0.0085)
$\mathbb{1}\{t = 2009\}:Z_2$	-0.0004 (0.0071)	0.0005 (0.0061)	0.0045 (0.0092)	-0.0058 (0.0085)
$\mathbb{1}\{t = 2011\}:Z_2$	-0.0214*** (0.0070)	-0.0085 (0.0061)	0.0095 (0.0094)	0.0124 (0.0087)
$\mathbb{1}\{t = 2012\}:Z_2$	-0.0379*** (0.0069)	-0.0206*** (0.0061)	0.0078 (0.0094)	0.0112 (0.0087)
$\mathbb{1}\{t = 2013\}:Z_2$	-0.0351*** (0.0068)	-0.0188*** (0.0060)	0.0080 (0.0094)	0.0027 (0.0087)
$\mathbb{1}\{t = 2008\}:Z_3$	0.0102 (0.0080)	0.0067 (0.0070)	0.0129 (0.0093)	0.0196** (0.0087)
$\mathbb{1}\{t = 2009\}:Z_3$	0.0029 (0.0081)	0.0020 (0.0071)	0.0100 (0.0093)	0.0127 (0.0088)
$\mathbb{1}\{t = 2011\}:Z_3$	-0.0657*** (0.0080)	-0.0323*** (0.0070)	0.0011 (0.0095)	0.0109 (0.0089)
$\mathbb{1}\{t = 2012\}:Z_3$	-0.0858*** (0.0080)	-0.0401*** (0.0070)	0.0007 (0.0097)	0.0012 (0.0090)
$\mathbb{1}\{t = 2013\}:Z_3$	-0.1059*** (0.0077)	-0.0539*** (0.0068)	0.0021 (0.0096)	0.0027 (0.0090)
R-squared	0.033	0.016	0.002	0.001
N	199488	199488	199488	199488

S3 Parameter estimates

In the tables below I use the original abbreviations to refer to different schooling programs. HAVO refers to applied high school, HBO to applied university, MBO3 and MBO4 to different levels of vocational education, and VMBO to pre-vocational secondary education.

Table S5: Nonpecuniary returns to education

	HAVO	HBO	MBO4	MBO3
Constant	-128442 (10225) 25479	16108 (2386)	56254 (5510)	66309 (8282)
G_2	(3670) 71296	—	—	—
G_3	(9390) 132759	—	—	—
G_4	(12529) 8598	—	—	—
θ_2	(1166) -8928	34150 (4764)	-43706 (6178)	-37411 (5265)
θ_3	(1369) -4773	-61922 (8950)	-47766 (5200)	60301 (7091)
θ_4	(750) 4402	38325 (6237)	27832 (4001)	50545 (7257)
θ_5	(642) 1960	7115 (1348)	7169 (1120)	-6911 (919)
Y_2	(284) -4792	10343 (1720)	3499 (536)	-19886 (3089)
Y_3	(607) -1293	4389 (701)	-1063 (166)	-2779 (267)
Y_4	(196)	11345 (2362)	462 (78)	-29778 (4550)

Table S6: Nonpecuniary returns to working

	Academic	Vocational
Age	-1157 (149)	3038 (488)
Constant	118755 (4199)	33900 (3809)
Y_2	14475 (1833)	14156 (1997)
Y_3	17893 (2863)	18354 (1945)
Y_4	35539 (4002)	30463 (3372)
MBO3	—	27171 (2604)
MBO4	—	38503 (3021)
VMBO	—	-3883 (681)

Table S7: Wage returns

	Academic	Vocational
	0.033	0.018
Age	(0.011)	(0.015)
	-0.002	-0.008
Age $\times I[Age > 30]$	(0.005)	(0.009)
	2.571	2.650
Constant	(0.111)	(0.155)
	0.056	0.027
Experience	(0.007)	(0.010)
	-0.177	-0.118
Experience ²	(0.028)	(0.025)
	0.017	0.031
Experience $\times \theta_2$	(0.014)	(0.010)
	0.038	0.025
Experience $\times \theta_3$	(0.014)	(0.008)
	0.001	0.007
Experience $\times \theta_4$	(0.015)	(0.009)
	0.010	-0.002
Experience $\times \theta_5$	(0.026)	(0.017)
	0.015	0.018
G_2	(0.023)	(0.019)
	0.017	0.016
G_3	(0.019)	(0.027)
	0.003	0.006
G_4	(0.029)	(0.033)
	0.141	-0.038
θ_2	(0.030)	(0.031)
	-0.018	-0.073
θ_3	(0.030)	(0.035)
	0.135	0.095
θ_4	(0.039)	(0.033)
	0.019	0.018
θ_5	(0.035)	(0.034)
		-0.009
Experience $\times$ MBO3	—	(0.007)
		-0.002
Experience $\times$ MBO4	—	(0.007)
		-0.004
Experience $\times$ VMBO	—	(0.009)
		0.035
Experience ² $\times$ MBO3	—	(0.024)
		0.027
Experience ² $\times$ MBO4	—	(0.024)
		0.010
Experience ² $\times$ VMBO	—	(0.025)
		0.181
MBO3	—	(0.038)
		0.164
MBO4	—	(0.026)
		0.097
Dropout	—	(0.029)
		0.029
VMBO	—	(0.026)

Table S8: Dropout Risk

	HBO	MBO4	MBO3	HAVO
Constant	0.436 (0.062)	0.891 (0.090)	0.921 (0.075)	0.262 (0.032)
G_2	0.224 (0.032)	0.136 (0.043)	—	0.323 (0.047)
G_3	0.490 (0.060)	0.184 (0.039)	—	0.610 (0.097)
G_4	0.958 (0.121)	0.217 (0.029)	—	0.906 (0.097)
θ_2	-0.046 (0.031)	—	—	-0.030 (0.029)
θ_3	-0.167 (0.036)	—	—	0.190 (0.036)
θ_4	-0.015 (0.030)	—	—	0.229 (0.034)
θ_5	-0.008 (0.029)	—	—	0.022 (0.030)
Y_2	0.113 (0.030)	0.215 (0.039)	0.051 (0.028)	0.034 (0.026)
Y_3	0.157 (0.034)	0.397 (0.056)	0.316 (0.045)	-0.014 (0.025)
Y_4	0.256 (0.043)	0.356 (0.058)	0.274 (0.047)	-0.008 (0.037)
MBO4	-0.251 (0.032)	—	—	—

Table S9: Degree Duration

	HBO	MBO4	MBO3
Constant	3.897 (0.452)	2.437 (0.391)	1.279 (0.144)
Y_2	-0.082 (0.037)	-0.047 (0.036)	-0.179 (0.038)
Y_3	-0.227 (0.040)	-0.011 (0.030)	-0.192 (0.032)
Y_4	-0.126 (0.030)	-0.026 (0.037)	-0.041 (0.026)
G_2	0.023 (0.041)	-0.121 (0.037)	—
G_3	-0.003 (0.037)	0.006 (0.034)	—
G_4	-0.195 (0.034)	-0.314 (0.047)	—

Table S10: Type probabilities

	Type 2	Type 3	Type 4	Type 5
Constant	0.378 (0.070)	1.381 (0.170)	-0.309 (0.048)	-0.269 (0.037)
G_2	0.582 (0.079)	-0.859 (0.140)	0.025 (0.035)	-0.017 (0.032)
G_3	0.118 (0.036)	-1.300 (0.176)	-0.146 (0.030)	-0.177 (0.024)
G_4	0.749 (0.121)	-1.655 (0.247)	0.990 (0.133)	-0.031 (0.029)
Z_2	-0.084 (0.028)	-0.166 (0.032)	0.346 (0.069)	-0.068 (0.040)
Z_3	-1.047 (0.127)	0.020 (0.042)	0.461 (0.082)	-0.287 (0.037)
Z_4	-0.123 (0.029)	0.108 (0.031)	-0.406 (0.064)	0.263 (0.042)
Y_2	-1.071 (0.170)	0.230 (0.036)	0.057 (0.032)	-0.258 (0.032)
Y_3	0.392 (0.068)	-0.292 (0.046)	1.403 (0.208)	-0.209 (0.032)
Y_4	-0.200 (0.033)	-0.238 (0.039)	0.794 (0.107)	0.131 (0.032)

Table S11: Transition costs high school

Transition Cost HAVO	
Z_2	70229.429 (8871.720)
Z_3	136365.636 (12086.435)
Z_4	147881.946 (13002.351)
Reform $\times Z_3$	-60000.000 (57805.061)
Reform $\times Z_4$	-160000.000 (73649.952)

Table S12: Distribution taste shocks

EV Shocks	
Scale	93225 (9674)

S4 Model fit

Table S13: Degree combinations by income

Income Quartile	Degree Combination	Observed	Estimated
Y_1	havo	0.038	0.041
	<i>havo – bachelor</i>	0.124	0.116
	mbo3	0.126	0.144
	<i>mbo3 – mbo4</i>	0.088	0.091
	<i>mbo3 – mbo4 – bachelor</i>	0.032	0.034
	mbo4	0.272	0.277
	vmbo	0.127	0.110
Y_2	havo	0.034	0.034
	<i>havo – bachelor</i>	0.120	0.116
	mbo3	0.132	0.127
	<i>mbo3 – mbo4</i>	0.091	0.099
	<i>mbo3 – mbo4 – bachelor</i>	0.033	0.031
	mbo4	0.283	0.300
	vmbo	0.085	0.099
Y_3	havo	0.031	0.034
	<i>havo – bachelor</i>	0.135	0.112
	mbo3	0.118	0.127
	<i>mbo3 – mbo4</i>	0.083	0.083
	<i>mbo3 – mbo4 – bachelor</i>	0.034	0.040
	mbo4	0.276	0.297
	vmbo	0.077	0.072
Y_4	havo	0.039	0.040
	<i>havo – bachelor</i>	0.174	0.160
	mbo3	0.099	0.111
	<i>mbo3 – mbo4</i>	0.071	0.074
	<i>mbo3 – mbo4 – bachelor</i>	0.032	0.035
	mbo4	0.261	0.271
	vmbo	0.080	0.065

Note: Each row reports the observed share of vocational school graduates with the given final degree combination, for the corresponding parental income quartile ("Observed"), compared to the model's simulated share at the estimated parameters ("Estimated").

Table S14: Degree combinations by vocational school GPA

Vocational School GPA Quartile	Degree Combination	Observed	Estimated
G_1	havo	0.006	0.011
	<i>havo – bachelor</i>	0.023	0.019
	mbo3	0.191	0.180
	<i>mbo3 – mbo4</i>	0.118	0.143
	<i>mbo3 – mbo4 – bachelor</i>	0.032	0.035
	mbo4	0.311	0.334
	vmbo	0.149	0.127
G_2	havo	0.019	0.025
	<i>havo – bachelor</i>	0.068	0.049
	mbo3	0.138	0.137
	<i>mbo3 – mbo4</i>	0.095	0.093
	<i>mbo3 – mbo4 – bachelor</i>	0.036	0.038
	mbo4	0.307	0.338
	vmbo	0.100	0.097
G_3	havo	0.040	0.042
	<i>havo – bachelor</i>	0.141	0.119
	mbo3	0.097	0.121
	<i>mbo3 – mbo4</i>	0.077	0.076
	<i>mbo3 – mbo4 – bachelor</i>	0.037	0.036
	mbo4	0.279	0.291
	vmbo	0.074	0.081
G_4	havo	0.078	0.073
	<i>havo – bachelor</i>	0.319	0.322
	mbo3	0.052	0.073
	<i>mbo3 – mbo4</i>	0.046	0.035
	<i>mbo3 – mbo4 – bachelor</i>	0.026	0.030
	mbo4	0.195	0.178
	vmbo	0.047	0.048

Note: Each row reports the observed share of vocational school graduates with the given final degree combination, for the corresponding vocational school GPA quartile ("Observed"), compared to the model's simulated share at the estimated parameters ("Estimated").

Table S15: Enrollment proportions by school type and vocational school GPA

Programme	School Type	Vocational School GPA Quartile	Observed	Estimated	
havo	Z_1	G_1	0.032	0.022	
		G_2	0.063	0.054	
		G_3	0.122	0.125	
		G_4	0.287	0.360	
	Z_2	G_1	0.046	0.041	
		G_2	0.099	0.095	
		G_3	0.206	0.202	
		G_4	0.436	0.469	
	Z_3	G_1	0.059	0.056	
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Programme	School Type	Vocational School GPA Quartile	Observed	Estimated
hbo	Z_4	G_2	0.148	0.117
		G_3	0.286	0.244
		G_4	0.502	0.552
		G_1	0.139	0.087
	Z_3	G_2	0.291	0.188
		G_3	0.426	0.350
		G_4	0.612	0.683
		G_1	0.319	0.348
	Z_2	G_2	0.434	0.473
		G_3	0.515	0.516
		G_4	0.655	0.695
		G_1	0.355	0.384
	Z_1	G_2	0.460	0.513
		G_3	0.550	0.572
		G_4	0.701	0.740
		G_1	0.350	0.355
	Z_4	G_2	0.469	0.484
		G_3	0.574	0.563
		G_4	0.720	0.766
		G_1	0.381	0.345
mbo3	Z_4	G_2	0.516	0.500
		G_3	0.616	0.574
		G_4	0.744	0.780
		G_1	0.508	0.490
	Z_3	G_2	0.421	0.385
		G_3	0.339	0.342
		G_4	0.212	0.223
		G_1	0.490	0.468
	Z_2	G_2	0.397	0.347
		G_3	0.307	0.304
		G_4	0.190	0.194
		G_1	0.487	0.489
	Z_1	G_2	0.381	0.370
		G_3	0.293	0.309
		G_4	0.179	0.165
		G_1	0.439	0.492
mbo4	Z_4	G_2	0.323	0.357
		G_3	0.251	0.297
		G_4	0.136	0.151
		G_1	0.831	0.859
	Z_3	G_2	0.861	0.889
		G_3	0.849	0.842
		G_4	0.726	0.688
		G_1	0.840	0.860
	Z_2	G_2	0.857	0.874
		G_3	0.813	0.804
		G_4	0.603	0.615
		G_1	0.838	0.845
	Z_1	G_2	0.836	0.842
		G_3	0.753	0.764
		G_4	0.549	0.544
		G_1	0.816	0.817
	Z_4	G_2	0.756	0.799
		G_3	0.648	0.688
		G_4	0.458	0.446
		G_1		

Note: Each row reports the observed enrollment proportion in a given program (Programme), for the corresponding school type and vocational school GPA quartile ("Observed"), compared to the model's simulated proportion at the estimated parameters ("Estimated").

Table S17: Enrollment proportions by income

Programme	Income Quartile	Observed	Estimated
havo	Y_1	0.231	0.220
	Y_2	0.211	0.200
	Y_3	0.221	0.201
	Y_4	0.268	0.277
hbo	Y_1	0.485	0.511
	Y_2	0.499	0.488
	Y_3	0.537	0.556
	Y_4	0.573	0.604
mbo3	Y_1	0.358	0.379
	Y_2	0.355	0.356
	Y_3	0.332	0.321
	Y_4	0.290	0.285
mbo4	Y_1	0.753	0.759
	Y_2	0.771	0.768
	Y_3	0.763	0.789
	Y_4	0.726	0.743

Note: Each row reports the observed enrollment proportion in a given program (Programme), for the corresponding parental income quartile ("Observed"), compared to the model's simulated proportion at the estimated parameters ("Estimated").

Table S18: Wage quartiles by final degree, age 30

Degree Type	Percentile	Observed	Estimated
bachelor	25%	20.654	21.774
	50%	24.312	24.766
	75%	28.184	28.162
havo	25%	16.254	17.755
	50%	19.481	20.374
	75%	23.833	23.345
mbo3	25%	16.418	17.777
	50%	19.183	20.246
	75%	22.568	23.003
mbo4	25%	17.221	18.781
	50%	20.299	21.491
	75%	23.970	24.530
vmbo	25%	15.063	16.015
	50%	17.890	18.394
	75%	21.506	21.018

Note: Each row reports the observed 25th, 50th, and 75th percentile of hourly wages at age 30 ("Observed") for full-time individuals by final degree, compared to the model's simulated percentile at the estimated parameters ("Estimated").

Table S19: Wage quartiles by final degree, age 35

Degree Type	Percentile	Observed	Estimated
bachelor	25%	23.870	25.870
	50%	28.887	29.796
	75%	34.372	34.374
havo	25%	17.802	18.598
	50%	21.923	22.561
	75%	27.441	26.358
mbo3	25%	18.142	19.013
	50%	21.368	22.003
	75%	25.012	25.327
mbo4	25%	19.232	20.233
	50%	23.122	23.649
	75%	27.899	27.516
vmbo	25%	16.120	16.678
	50%	19.580	19.447
	75%	24.397	22.543

Note: Each row reports the observed 25th, 50th, and 75th percentile of hourly wages at age 35 ("Observed") for full-time individuals by final degree, compared to the model's simulated percentile at the estimated parameters ("Estimated").

Table S20: Wage quartiles by final degree, age 40

Degree Type	Percentile	Observed	Estimated
bachelor	25%	26.708	27.868
	50%	33.028	32.686
	75%	39.874	38.827
havo	25%	18.213	18.234
	50%	22.926	23.260
	75%	29.292	28.256
mbo3	25%	18.923	19.271
	50%	22.275	22.965
	75%	26.485	26.989
mbo4	25%	20.742	20.531
	50%	25.157	24.950
	75%	30.761	29.762
vmbo	25%	16.333	16.367
	50%	19.843	19.713
	75%	24.588	23.205

Note: Each row reports the observed 25th, 50th, and 75th percentile of hourly wages at age 40 ("Observed") for full-time individuals by final degree, compared to the model's simulated percentile at the estimated parameters ("Estimated").

Table S21: Wage equation no bachelor's degree

	Observed	Estimated
Coefficients		
Intercept	2.552	2.684
Experience	0.025	0.041
Experience ²	-0.000	-0.001
G_2	0.012	0.022
G_3	0.017	0.005
G_4	0.027	0.009
Y_2	0.012	-0.017
Y_3	0.026	0.012
Y_4	0.042	-0.005
mbo3	0.072	0.046
Experience $\times$ mbo3	-0.008	-0.002
mbo4	0.071	0.095
Experience $\times$ mbo4	-0.004	-0.001
RSE	0.238	0.212
vmbo	-0.013	-0.110
Experience $\times$ vmbo	-0.006	-0.002

Note: Coefficients of the auxiliary vocational wage regression described in the main text, estimated on full-time individuals who left school without a bachelor's degree, comparing the observed coefficients ("Observed") to the model's simulated fit at the estimated parameters ("Estimated").

Table S22: Wage equation bachelor's degree holder

	Observed	Estimated
Coefficients		
Intercept	2.713	2.746
Experience	0.067	0.065
Experience ²	-0.003	-0.002
G_2	-0.000	0.018
G_3	0.004	0.005
G_4	0.012	-0.013
Y_2	-0.003	-0.021
Y_3	0.002	0.004
Y_4	0.010	-0.010
$mbo3 - mbo4 - bachelor$	-0.007	0.057
$mbo4 - bachelor$	0.008	0.020
RSE	0.220	0.202
T^u	0.016	0.031

Note: Coefficients of the auxiliary academic wage regression described in the main text, estimated on full-time individuals who left school with a bachelor's degree, comparing the observed coefficients ("Observed") to the model's simulated fit at the estimated parameters ("Estimated").

Table S23: Wages by school type: vocational path

	Observed	Estimated
Coefficient		
Intercept	3.023	2.992
G_2	0.035	0.031
G_3	0.054	0.023
G_4	0.081	0.043
Z_2	-0.006	0.005
Z_3	-0.008	-0.002
Z_4	-0.011	-0.010

Note: Coefficients of a regression of log wages on school type and vocational school GPA fixed effects (described in the main text), estimated on individuals who did not enroll in applied high school, comparing the observed coefficients ("Observed") to the model's simulated fit at the estimated parameters ("Estimated").

Table S24: Wages by school type: alternative path to applied university

	Observed	Estimated
Coefficient		
Intercept	3.169	3.074
G_2	0.009	0.024
G_3	0.013	0.015
G_4	0.029	0.019
Z_2	-0.013	0.001
Z_3	-0.016	-0.008
Z_4	-0.010	-0.010

Note: Coefficients of a regression of log wages on school type and vocational school GPA fixed effects (described in the main text), estimated on individuals who did not enroll in applied high school and who hold a bachelor's degree - i.e., who reached applied university via the alternative path - comparing the observed coefficients ("Observed") to the model's simulated fit at the estimated parameters ("Estimated").

Table S25: Wages by school type: direct path

	Observed	Estimated
Coefficient		
Intercept	3.073	3.054
G_2	0.028	0.048
G_3	0.051	0.045
G_4	0.084	0.052
Z_2	-0.005	-0.003
Z_3	-0.008	-0.030
Z_4	-0.010	-0.018

Note: Coefficients of a regression of log wages on school type and vocational school GPA fixed effects (described in the main text), estimated on individuals who enrolled in applied high school, comparing the observed coefficients ("Observed") to the model's simulated fit at the estimated parameters ("Estimated").

Table S26: Wages by school type: full sample

	Observed	Estimated
Coefficient		
Intercept	3.023	2.990
G_2	0.038	0.036
G_3	0.062	0.037
G_4	0.100	0.071
Z_2	-0.002	0.008
Z_3	-0.002	-0.001
Z_4	0.000	0.001

Note: Coefficients of a regression of log wages on school type and vocational school GPA fixed effects (described in the main text), estimated on the full sample of full-time individuals, comparing the observed coefficients ("Observed") to the model's simulated fit at the estimated parameters ("Estimated").

Table S27: Applied university enrollment by school type: direct path

	Observed	Estimated
Coefficient		
Intercept	0.578	0.731
G_2	0.118	0.080
G_3	0.196	0.110
G_4	0.291	0.159
Z_2	0.009	-0.004
Z_3	0.010	-0.005
Z_4	0.020	-0.002

Note: Coefficients of a regression of applied university enrollment on school type and vocational school GPA fixed effects, estimated on individuals who enrolled in applied high school, comparing the observed coefficients ("Observed") to the model's simulated fit at the estimated parameters ("Estimated").

Table S28: Applied university graduation by school type: direct path

	Observed	Estimated
Coefficient		
Intercept	0.422	0.443
G_2	0.104	0.080
G_3	0.167	0.165
G_4	0.267	0.277
Z_2	0.003	0.003
Z_3	-0.001	0.003
Z_4	0.003	0.000

Note: Coefficients of a regression of bachelor's degree completion on school type and vocational school GPA fixed effects, estimated on individuals who enrolled in applied high school, comparing the observed coefficients ("Observed") to the model's simulated fit at the estimated parameters ("Estimated").

Table S29: Applied university enrollment by school type: alternative path

School Type	Vocational School GPA Quartile	Observed	Estimated
Z_1	G_1	0.312	0.339
	G_2	0.418	0.455
	G_3	0.480	0.469
	G_4	0.564	0.584
Z_2	G_1	0.344	0.369
	G_2	0.432	0.484
	G_3	0.493	0.503
	G_4	0.559	0.613
Z_3	G_1	0.336	0.335
	G_2	0.430	0.442
	G_3	0.489	0.476
	G_4	0.557	0.612
Z_4	G_1	0.344	0.308
	G_2	0.431	0.425
	G_3	0.479	0.432
	G_4	0.537	0.555

Note: Each row reports the observed proportion of individuals who enroll in applied university ("Observed"), by school type and vocational school GPA quartile, among those who did not enroll in applied high school, compared to the model's simulated proportion at the estimated parameters ("Estimated"). Corresponds to the alternative-path population shown in the "Variation across school types" figure in the main text.

Table S30: Applied university graduation by school type: alternative path

School Type	Vocational School GPA Quartile	Observed	Estimated
Z_1	G_1	0.185	0.197
	G_2	0.284	0.282
	G_3	0.358	0.321
	G_4	0.452	0.457
Z_2	G_1	0.209	0.205
	G_2	0.286	0.302
	G_3	0.355	0.342
	G_4	0.445	0.482
Z_3	G_1	0.200	0.187
	G_2	0.293	0.277
	G_3	0.351	0.326
	G_4	0.430	0.474
Z_4	G_1	0.204	0.175
	G_2	0.289	0.268
	G_3	0.344	0.301
	G_4	0.421	0.422

Note: Each row reports the observed proportion of individuals who complete a bachelor's degree ("Observed"), by school type and vocational school GPA quartile, among those who did not enroll in applied high school, compared to the model's simulated proportion at the estimated parameters ("Estimated"). Corresponds to the alternative-path population shown in the "Variation across school types" figure in the main text.

Table S31: Model fit: difference-in-differences moments

Outcome	School Type	Observed	Simulated	Difference
First Stage (HAVO Enrollment)	2	-0.0337	-0.0395	0.0059
	3	-0.0906	-0.0915	0.0010
HAVO Degree	2	-0.0180	-0.0188	0.0008
	3	-0.0450	-0.0528	0.0077
HBO Enrollment	2	0.0041	0.0031	0.0010
	3	-0.0064	-0.0077	0.0013
Bachelor Degree	2	0.0076	0.0047	0.0028
	3	-0.0057	-0.0045	-0.0013

Note: Each row reports the estimated effect of the reform for the given outcome and school type (relative to Z_1/Z_2 pooled as control), comparing the value targeted in estimation ("Observed") to the model's fit at the estimated parameters ("Simulated").

Table S32: Model fit: untargeted difference-in-differences wage effect

Outcome	School Type	Observed	Simulated	Difference
Log Wage	2	0.0074	-0.0004	0.0078
	3	0.0057	0.0021	0.0037

Note: The wage effect of the reform is not targeted in estimation (the estimated effect is not statistically significant; see the "Exam rule reform" subsection in the main text). This table compares it to the model's untargeted implied wage effect at the estimated parameters.